

Optimization – 4

Discrete optimisation: rearrangement, exchange and greed

Homework

10 сентября 2026 года

Theoretical minimum. The rearrangement inequality and Abel summation; Chebyshev's sum inequality and Muirhead; majorization, Schur convexity and Schur's criterion; the exchange and smoothing arguments; the greedy algorithm and matroids.

1. Prove Schur's criterion for Schur convexity and deduce Muirhead's inequality from it. Then find the maximum and the minimum of $\sum x_i^3$ given $\sum x_i$ and $\sum x_i^2$ for non-negative x_i .
2. (IMC) Let k be a positive integer. Find the smallest n for which there exist non-zero vectors $v_1, \dots, v_k$ in $\mathbb{R}^n$ such that $v_i \perp v_j$ whenever $|i - j| > 1$.
3. (MIPT) Every two of $n \geq 4$ cities are joined by a one-way road, and one can get from any city to any other. Find the smallest k such that for any three cities A, B, C one can travel from A to B and then from B to C using at most k roads in total, a road used twice being counted twice.
4. (OMOUS 2023) There are 2023 displays and 2023 remotes, and we do not know which remote belongs to which display. In one operation we may pick any set of remotes; all of them switch between off and on simultaneously, and we see which displays light up. Find the least number of operations needed to match the remotes to the displays.
5. (IMS 2024) Ayla has 100 distinct real numbers and Bardia wants to find the index of the smallest. At each step Bardia names two indices and Ayla says which of the two numbers is smaller; Ayla may lie at

most once. Determine the smallest number of questions guaranteeing success.

6. (Высшая лига 2012) Find the smallest number of proper subspaces of $\mathbb{F}_2^3$ whose union is the whole space.
7. (IMS 2008) A subset of an $n \times n$ table is called even if it contains an even number of cells in every row and in every column. Find the smallest k such that every subset of k cells contains a non-empty even subset.
8. (IMC 2025 shortlist) Let n be a positive integer. Find the least positive integer k such that for every complex $n \times n$ matrix A at least one of $A, A + I_n, A + 2I_n, \dots, A + (k - 1)I_n$ is invertible.
9. Prove that among all non-negative sequences with a given sum and a given number of terms the sum of squares is minimal for the constant sequence and maximal for the most unbalanced one, and restate both facts through majorization.
10. (Сибирская 2022) Dima writes a natural number in binary, $N = a_0 + a_1 2 + \dots + a_k 2^k$, and Tanya reads the same digits as a ternary number $M = a_0 + a_1 3 + \dots + a_k 3^k$.
 - a) Prove that $c_1 M^{\log_3 2} \leq N \leq c_2 M^{\log_3 2}$ for some constants independent of N .
 - b) Find the largest such c_1 and the smallest such c_2 .